Aryan Golbaghi

SOFTWARE & AI ENGINEER

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PROFESSIONAL PROFILE

Software and AI engineer with hands-on ownership of Python applications, retrieval-augmented generation (RAG) systems and data pipelines used by executives and investment teams. Combines software engineering, LLM evaluation and model fine-tuning with experience in developer tooling and real-time observability. First-Class MComp Computer Science (Artificial Intelligence) graduate from the University of Sheffield in 2026, and first author of an arXiv machine-learning preprint.

TECHNICAL SKILLS

Programming: Python, SQL
AI / ML: PyTorch, NumPy, RAG, BM25, LLM evaluation, model fine-tuning, Langfuse, DeepEval
Backend / Data: FastAPI, pandas, REST APIs, SQLite, Qdrant
Cloud / DevOps: AWS, Microsoft Azure, Amazon Bedrock, GitHub Actions, CircleCI, HPC, SLURM, CUDA, DeepSpeed, Mamba
Engineering: pytest, TDD, Agile/Scrum, Docker, Linux

PROFESSIONAL EXPERIENCE

Lendable **Jun 2026 – Present**

AI & Infrastructure Analyst | London, UK

- Developed a Python/FastAPI platform for AI-assisted macroeconomic research across 37 markets, integrating Bloomberg, IMF and World Bank data; adopted by the CEO and saving approximately half a day to one day of manual data collection and review.
- Co-designed and implemented a reusable RAG document assistant adopted across investment, marketing and other teams, with one-time ingestion, OCR and table-aware chunking, saving hours of manual document retrieval.
- Engineered entity- and period-aware BM25 retrieval, numerical fact verification and abstention controls; achieved an 86.9% evidence-page hit rate on 130 answerable held-out questions, supported by repeatable evaluations and regression tests.
- Owned development of a Python/pandas portfolio pipeline used by Risk, ALM and Portfolio Analysis, automating reporting for 250+ live loan positions and replacing recurring manual preparation with a single-command workflow.
- Built source archives, effective-dated configuration and 125+ automated verification checks to make reporting reproducible and prevent failed runs from replacing the current report pack.
- Automated loan valuation and reconciliation for a six-person Risk team, saving approximately half a day of recurring monthly work.

Redgate Software **Jun 2025 – Sep 2025**

Software Engineer Intern | Cambridge, UK

- Worked on a command-line utility that generated undo scripts for Flyway migrations, reducing rollback preparation time and improving developer safety.
- Delivered Flyway Desktop features using test-driven development; replaced deprecated third-party UI components and updated associated tests.
- Conducted an AI feasibility study by interviewing all Flyway Desktop tech leads, prioritising use cases, evaluating technical risk and ROI, and presenting recommendations to the team.

SportLight **Oct 2024 – Jun 2026**

Part-time Software Engineer | Bicester, UK

- Developed Python features and operational tooling for real-time observability, including metrics, logging, alerting and failure handling, reducing on-site incidents.
- Designed and implemented a real-time failure-management system that cut recovery time by three hours per failure.
- Managed and mentored four monitors supporting live games, guiding effective responses to real-time incidents.
- Collaborated with product and engineering teams to triage issues, ship fixes and write runbooks that prevented repeat faults.

Ockana **Jun 2024 – Sep 2024**

AI Software Engineer | London, UK

- Refactored a PDF automation pipeline from a monolithic class into an extensible subclass-based architecture, automating document parsing and processing while reducing manual intervention.
- Built an LLM monitoring and testing framework with Langfuse and AutoRAG; defined evaluation methods to measure output accuracy and improve transparency for stakeholders.
- Developed a synthetic-data generator using DeepEval to evaluate and optimise RAG models.
- Presented LLM solutions and evaluation findings to non-technical stakeholders, translating model behaviour into practical guidance.

SELECTED RESEARCH & PROJECTS

Beyond AUC Credit Risk Modelling and Validation

AI-assisted independent research | Technical report | 2026

- Built a reproducible Python study on 1,000 public credit records with nested cross-validation, sigmoid calibration and 2,000 paired bootstrap replicates.
- Measured holdout AUCs of 0.790 for boosting and 0.749 for logistic regression; found unstable fold rankings and no convincing calibration benefit.

Enhancing Speech Emotion Recognition via Fine-Tuning Pre-Trained Models and Hyper-Parameter Optimisation

First Author | arXiv, 2025

- Engineered a resource-efficient AutoML/HPO workflow for speech emotion recognition using pre-trained representations, Bayesian optimisation and TPE.
- Achieved 0.97 balanced class accuracy on EmoDB using eight CPU cores; improved cross-lingual zero-shot accuracy by 0.25 on CREMA-D and 0.26 on RAVDESS.

DPO-Mini — Preference Training & Performance Evaluation

AI-assisted project | Technical report | 2026

- AI-assisted implementation of a PyTorch DPO pipeline with LoRA fine-tuning, automated correctness tests and TRL loss/gradient checks with documented FP32 limitations.
- Reference-score caching on SmolLM2-135M-Instruct cut median whole-process CPU benchmark time by 11.7% versus a shared-base reference over three trials per strategy, including cold-cache construction.

Model Alignment for Reliable LLM Abstention

Reinforcement Learning Lead | Manuscript in preparation

- Leading reinforcement-learning implementation for a multidisciplinary study of LLM uncertainty and refusal mechanisms using GRPO and DPO.
- Fine-tuned pre-trained LLM models on a university HPC cluster using GPU compute, managing batch jobs, resource allocation, training configurations and experiment outputs.
- Preliminary experiments increased recall by 19.2% with GRPO and 83.33% with DPO relative to prior baselines while investigating hallucination reduction and answer coverage.

EDUCATION

University of Sheffield **Sep 2022 – Jun 2026**

MComp Computer Science (Artificial Intelligence) – First-Class Honours

Selected results: Computer Vision 89%, Automata 80%, Speech Processing 78%, Robotics 78%, Natural Language Processing 77%.

City and Islington College **Sep 2020 – Aug 2022**

A Levels: Mathematics A, Further Mathematics A, Computer Science B*

LEADERSHIP & ADDITIONAL EXPERIENCE

Founding President, Kurdish Society **Sep 2025 – Present**

University of Sheffield | Sheffield, UK

- Founded an SU-recognised society, recruited the inaugural committee, grew membership to 30+ in the first month, delivered 5+ events and secured 50 sign-ups at the Activities Fair.

Secretary, Persian Society **Sep 2024 – Jun 2025**

University of Sheffield | Sheffield, UK

- Maintained governance records, scheduled committee meetings, produced action-focused minutes and tracked follow-ups to completion.

Kurdish/Persian Interpreter **Mar 2022 – Aug 2024**

UK Language Solutions | England

- Interpreted between Kurdish, Persian and English in high-stakes NHS, council, education, immigration and family-court settings, providing accurate communication and cultural mediation.